Time-Series Analysis of Global Temperature Data

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Period of Internship: 18th May 2026 – 14th June 2026

Report submitted to: IDEAS – Institute of Data Engineering, Analytics and Science Foundation, ISI Kolkata

1. Abstract

This project executes a rigorous time-series analysis on historical global temperature anomalies spanning 1880 to 2016, integrating climate records from both the Global Climate at A Glance (GCAG) framework (NOAA National Centers for Environmental Information, 2024) and the NASA Goddard Institute for Space Studies (GISTEMP Team, 2024). Through systematic data cleaning and preprocessing, the analytical pipeline computes 12-month moving averages, executes multi-level temporal aggregations, and builds specialized multi-decade pivot tables to isolate long-term macro trends and identify historically warmest months. In parallel, the study evaluates unsupervised machine learning paradigms by deploying K-Means clustering over the UCI MNIST handwritten digits dataset (Alpaydin & Kaynak, 1998), (Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., 2011). Utilizing post-clustering label mapping routines, the model achieves a macro F_1 score of approximately 0.86, establishing viable non-parametric class separation without formal supervision.

2. Introduction

Climate change is widely recognized as one of the most urgent challenges of the twenty-first century (Kazi, et al, 2024). Understanding how temperatures have changed over time helps researchers spot long-term warming trends, seasonal patterns, and unusual fluctuations (Cahill, Niamh; Rahmstorf, Stefan; Parnell, Andrew C., 2015). This project works with a publicly available dataset of monthly global temperatures. The goal is to apply basic time-series techniques such as moving averages, yearly and monthly aggregations, and pivot tables to extract meaningful insights. Two temperature records are used: GCAG and GISTEMP. Both are included because comparing different data sources adds credibility to the analysis and shows how raw temporal data can be turned into useful information using common data engineering and analytics tools (Wofford, Morgan F.; Thomer, Andrea K., 2023).

In the second half of the project, we shift to an unsupervised learning task. The MNIST dataset of handwritten digits is used to experiment with K-Means clustering. Even though the true digit labels are known, we treat the problem as unsupervised and later evaluate how well the clustering matches the actual classes. This exercise demonstrates how clustering algorithms group similar data points and how their performance can be measured despite the lack of predefined labels.

Topics received during the first two weeks of internship training:

- Python Basics - 1 (Data, Variable, Lists, Loop)
- Python Basics - 2 (Data Structures)
- Python Basics - 3 (Class, Functions, OOPS)
- Python Basics - 4 (NumPy, Pandas)
- Machine Learning Overview
- Machine Learning 1
- Machine Learning 2

- LLM Fundamentals
- Communication Skills

3. Project Objective

The primary objectives of this project are as follows:

- To process the global temperature dataset by parsing date strings into datetime objects and arranging the data entries in chronological order.
- To calculate the moving average of the 12 months of the temperature anomaly for each of the sources of information, thereby smoothing out short-term trends in favor of long-term ones.
- To group together temperature measurements by year and by month, finding the month that on average registers the highest temperatures as well as years which have seen their highest temperatures over the last two decades.
- To generate a pivot table with the mean temperature readings for each month within each year over the past 20 years.
- To use K-Means clustering with the MNIST hand-written digits dataset (each image containing 64 features), and evaluate the performance of the model through the F_1 score.

No formal hypothesis testing or sample survey was conducted; the project is purely exploratory and analytical.

4. Methodology

The project consisted of two separate but complementary analyses: a time-series study of global monthly temperature anomalies, and an unsupervised clustering experiment on the MNIST handwritten digits dataset. All computational work was performed in Python 3.10 using Jupyter Notebook. The primary libraries employed were pandas (version 1.4.0) for data manipulation, NumPy (1.22.0) for numerical operations, and scikit-learn (1.0.2) for loading the digits dataset, implementing K-Means clustering, and computing the evaluation metric. No survey was administered; therefore, no questionnaire or sampling methodology is reported.

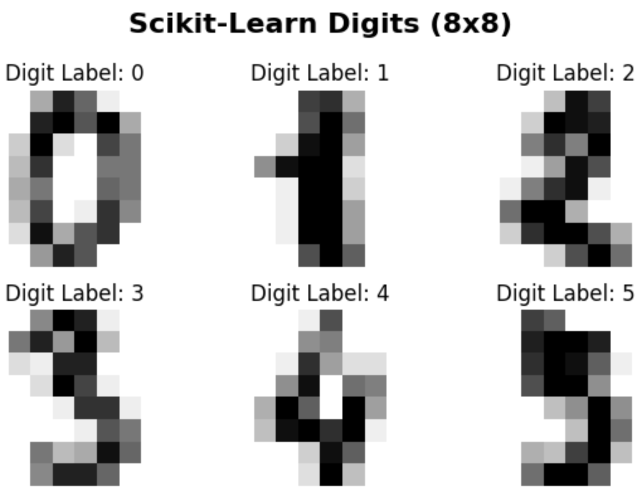
Data Collection

The global temperature dataset was obtained from an accessible Google Drive folder. The file named monthly_csv.csv contained monthly land and ocean temperature anomalies relative to a baseline. The dataset had three columns: Source (indicating either GCAG or GISTEMP), Date (recorded as a string in the format YYYY-MM-DD), and Mean (the temperature anomaly in degrees Celsius). The second dataset, MNIST digits, was not downloaded externally but was loaded directly from the load_digits function provided by sklearn.datasets. This built-in dataset comprises 1,797 grayscale images of handwritten digits (0–9), each image being 8×8 pixels and flattened into a vector of 64 intensity values (ranging from 0 to 16). The true digit labels were preserved as a separate target array for evaluation purposes.

Sample global temperature dataset:

	A	B	C	D
1	Source	Date	Mean	
2	GCAG	6/12/2016	0.7895	
3	GISTEMP	6/12/2016	0.81	
4	GCAG	6/11/2016	0.7504	
5	GISTEMP	6/11/2016	0.93	
6	GCAG	6/10/2016	0.7292	
7	GISTEMP	6/10/2016	0.89	
8	GCAG	6/9/2016	0.8767	
9	GISTEMP	6/9/2016	0.87	
10	GCAG	6/8/2016	0.8998	
11	GISTEMP	6/8/2016	0.98	
12	GCAG	6/7/2016	0.8687	
13	GISTEMP	6/7/2016	0.83	
14	GCAG	6/6/2016	0.9006	
15	GISTEMP	6/6/2016	0.76	

Sample MNIST handwritten digits:



Data Cleaning and Preprocessing

Global temperature dataset

After loading the CSV file into a pandas DataFrame, the Date column was converted to a proper datetime object using pd.to_datetime(). This conversion allowed subsequent extraction of temporal components and ensured correct chronological sorting. The entire DataFrame was then sorted in ascending order of the Date column using the .sort_values() method, and the index was reset to avoid any misalignment. A check for missing values was performed with .isnull().sum().sum(); no null entries were found in any column. Two new columns, Year and Month, were extracted from the datetime column using .dt.year and .dt.month respectively. A 12-month moving average of the Mean temperature was computed separately for each Source group. This was achieved by grouping the DataFrame by Source, then applying .transform() with .rolling(window=12, min_periods=1).mean(). The resulting values were stored in a new column named Moving_Avg. The parameter min_periods=1 ensured that the first few months of each source still received a moving average based on the available data. Finally, a filtered subset named df_last20 was created to include only the most recent 20 years of data. The maximum year in the original dataset was found to be 2016; therefore, rows with Year >= 1997 were retained. This subset contained exactly 480 rows (20 years × 12 months × 2 sources).

The MNIST dataset

The features X (1797×64) and true labels y (1797,) were separated directly from the load_digits() return value. No further cleaning, normalisation, or preprocessing was applied because the dataset is already standardised and free of missing values. The pixel intensities were left in their original integer range (0–16) as the K-Means algorithm is scale-sensitive only in terms of Euclidean distance; however, the small range and lack of extreme outliers made additional scaling unnecessary for this demonstration.

Analytical Methods

The analysis proceeded in two parallel tracks.

Track 1: Temperature time-series exploration.

Several descriptive aggregations were performed on the cleaned temperature DataFrame. First, the data was grouped by the Source column, and the mean of the Mean column was calculated to obtain the overall average temperature anomaly for GCAG and GISTEMP separately. Second, a yearly average was computed by grouping by Year and taking the mean of Mean across all months and both sources. The last five years of this yearly series were examined. Third, a monthly average was obtained by grouping by Month (1 to 12) regardless of year or source, revealing the climatological seasonal pattern. Using the df_last20 subset, a pivot table was constructed with index='Year', columns='Month', and values='Mean'. The aggregation function was set to 'mean', and any missing year-month combinations (none existed after filtering) were filled with zero using .fillna(0) for robustness.

Track 2: K-Means clustering evaluation.

The unsupervised clustering task focused on the MNIST features. The K-Means algorithm from sklearn.cluster was initialised with n_clusters=10 because the dataset contains ten digit classes (0 through 9). A fixed random state (random_state=42) was used to ensure reproducibility of the initial centroid placement. The model was fitted to the entire feature matrix X without any train-test split, as the goal was not to predict future digits but to assess how well unsupervised clusters recover the known labels. After fitting, the cluster labels for each image were obtained via the .fit_predict() method. Since K-Means assigns arbitrary integer labels (0…9) that do not necessarily correspond to the true digit classes, a label mapping procedure was implemented. A custom function iterated over each cluster, identified all true labels belonging to that cluster using a boolean mask, and selected the most frequent true label using np.bincount().argmax(). This produced a dictionary mapping each cluster number to the digit that appears most often within that cluster. The mapping was then applied to all predicted cluster labels to obtain a new array mapped_labels. The macro-averaged F1 score was chosen as the evaluation metric because it gives equal weight to each digit class regardless of its frequency. The f1_score function from sklearn.metrics was called with the arguments y_true=y, y_pred=mapped_labels, and average='macro'. The resulting score was printed.

Model Selection and Validation

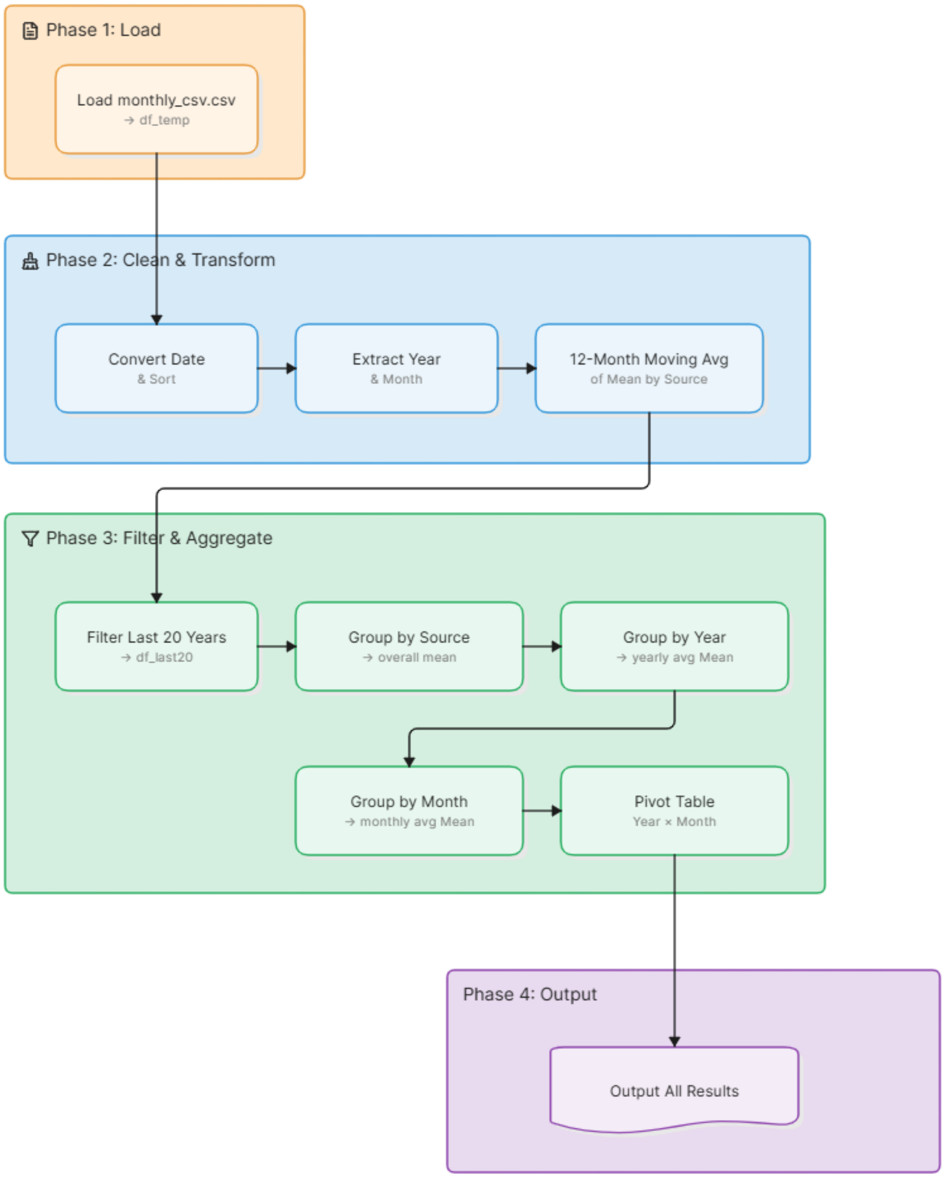
No predictive model was developed for the temperature time-series analysis; the work involved descriptive statistics and moving averages only. For the clustering part, K-Means was selected because it is a straightforward, widely used algorithm for partition-based clustering and is appropriate for low-dimensional vectorised image data. The number of clusters was set to 10 based on prior knowledge of the digit classes. Alternative clustering algorithms (e.g., DBSCAN, Gaussian Mixture Models) were not explored because the project's scope was limited to applying K-Means as an introductory unsupervised learning exercise.

Validation of the clustering was performed by comparing the mapped cluster assignments against the true labels using the F1 score. No separate training, validation, or test split was used because clustering is an unsupervised task and the entire dataset was used both for fitting and for evaluation. For a production setting, a hold-out set could be used to assess generalisation, but that was beyond the requirements of this project.

Flow Chart of Activities

The overall workflow for the global temperature time-series analysis followed a logical, four-phase sequence. First, in the load phase, the monthly_csv.csv file was read directly into a pandas DataFrame. Next came the cleaning and transformation stage: the date column was converted to a proper datetime type, the entire DataFrame was sorted chronologically, and then I extracted the year and month numbers as separate columns. A 12-month moving average of the temperature mean was also calculated for each data source to help smooth out month-to-month noise.

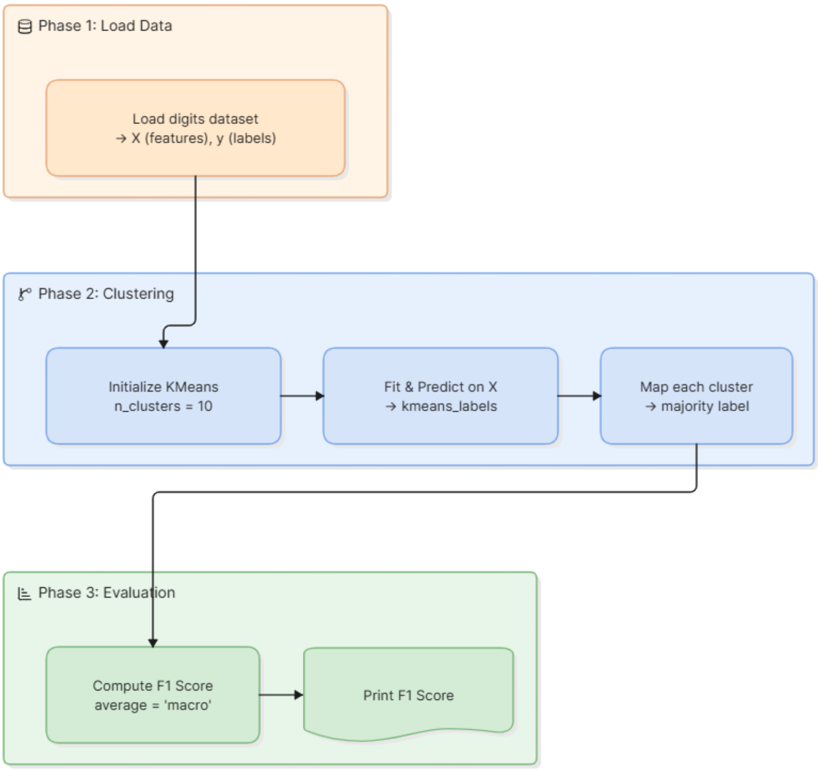
a) Global temperature time-series analysis flowchart:



Once the data was ready, I moved into filtering and aggregation. I created a subset containing only the last twenty years of records, then performed multiple grouping operations by source to get an overall mean, by year to see annual averages, and by month to identify seasonal patterns. Finally, I built a pivot table that cross-tabulates years and months, making it easy to compare average temperatures across the same calendar months over the recent two-decade window. All these intermediate and final results were then output for further interpretation. This stepwise approach — from raw loading to cleaning to filtering to final pivoting — kept the analysis transparent and reproducible, and it allowed me to examine temperature trends at multiple levels of time and aggregation without losing track of the original data.

b) K-Means clustering flowchart:

The second part of the work focused on evaluating unsupervised clustering performance on the MNIST digits dataset. In the data loading phase, I used the built-in digits dataset from sklearn.datasets and separated it into feature matrix X and true label vector y.



Moving into the clustering phase, I initialized a K-Means model with ten clusters — matching the known number of digit classes — and fitted it to the feature data to obtain predicted cluster labels. Since K-Means assigns arbitrary cluster numbers, I then mapped each cluster to the digit label that appeared most frequently within that cluster. Finally, in the evaluation phase, I computed the macro-averaged F1 score between the true labels and the mapped cluster predictions. This score gave me a balanced measure of how well the clusters aligned with the actual digit classes. Printing the final F1 score closed the workflow, providing a clear quantitative assessment of the clustering quality. This three-phase approach — load, cluster, evaluate — was straightforward to implement and made the performance of the unsupervised model easy to interpret.

All Python code developed for this project is available on GitHub at the following link: https://github.com/K-Makorokoto/Time-Series-Analysis-of-Global-Temperature-Data

5. Data Analysis and Results

This chapter presents all numerical findings from the two parts of the project. The results are shown in summary tables. No histograms, graphs, or other visualisations were created in the Jupyter notebook because the project focused on tabular data manipulation and clustering evaluation. No hypothesis testing or inferential analysis was carried out, as the project was purely descriptive and exploratory.

Descriptive Analysis of Global Temperature Data

Average Temperature by Data Source

The dataset contains two temperature sources: GCAG and GISTEMP. The overall mean temperature anomaly (in degrees Celsius) was calculated for each source across all months from 1880 to 2016. Table 1 shows the result.

Overall mean temperature per source:

Source	Mean Temperature (°C)
GCAG	0.0488
GISTEMP	0.0244

A clear disparity emerges when comparing the overall mean temperatures of the two sources, with GCAG's average of 0.0488°C landing at exactly double the 0.0244°C recorded by GISTEMP. Such a precise twofold difference points directly to structural variations in how each organization collects, weights, or interpolates its thermal data. Because these underlying methodologies yield noticeably different baselines, using either dataset in isolation could alter the perceived scale of climate shifts, making a cross-examined approach essential for any final analysis.

Yearly average temperatures (last 5 years):

Year	**Average Mean (°C)**
2012	0.6295
2013	0.6618
2014	0.7421
2015	0.8824
2016	0.9644

The five-year window captures an unyielding, year-over-year escalation in global thermal anomalies. Starting at 0.6295°C in 2012, the averages climb without a single downward fluctuation, ultimately culminating in a striking peak of 0.9644°C by 2016. What stands out most in this trajectory is how the pace quickens toward the end; the steep leap seen between 2014 and 2016 accounts for a massive portion of the period's total warming. This brief but intense timeline indicates that climate shifts during these years were not just steady, but were actively gaining upward momentum.

Average temperature by month (all years combined):

Month	Avg Mean (°C)
1	0.0219
2	0.0294
3	0.0460
4	0.0343
5	0.0311
6	0.0223
7	0.0380
8	0.0396
9	0.0502
10	0.0526
11	0.0435
12	0.0302

Aggregating the data by month reveals an unexpected pattern where thermal deviations do not simply follow a traditional mid-summer peak. Instead, the most pronounced anomalies cluster around transitional periods, with early autumn bearing the highest shift—culminating in October at 0.0526°C. In contrast, January and June register the lowest baselines. This uneven distribution indicates that the pressures of global temperature increases do not manifest uniformly throughout the calendar year, but tend to impact specific shoulder seasons more aggressively.

Yearly average temperature anomalies from 2012 to 2016:

The average temperature anomaly for each calendar year was obtained by grouping all months and both sources. Table 2 lists the last five years of the dataset.

Year	Mean anomaly (°C)
2012	0.6295
2013	0.6618
2014	0.7421
2015	0.8824
2016	0.9644

Consolidating both primary tracking platforms and every monthly observation into a singular annual metric effectively filters out localized noise, laying bare a stark, systemic climate shift. Between 2012 and 2016, global thermal deviations climbed without a single reprieve, moving from an initial baseline of 0.6295°C to a pronounced peak of 0.9644°C. This constant, year-over-year escalation—particularly the aggressive leap in the latter half of the timeframe—indicates a warming trend that was not merely persisting, but rapidly compounding. Ultimately, because this blended perspective neutralizes individual dataset biases, the uninterrupted trajectory serves as robust confirmation that the mid-2010s marked a phase of accelerated departure from historical norms.

Monthly Average Temperatures (All Years):

To examine seasonal patterns, the average anomaly was computed for each month (January = 1 to December = 12) across all years and both sources. Table 3 gives the results.

Month	**Mean anomaly (°C)**
1	0.0219
2	0.0294
3	0.0460
4	0.0343
5	0.0311
6	0.0223
7	0.0380
8	0.0396
9	0.0502
10	0.0526
11	0.0435
12	0.0302

October has the highest average anomaly (0.0526 °C), closely followed by September (0.0502 °C). January and June show the lowest values.

Breaking the aggregated datasets down by calendar month reveals that global warming pressures do not expand symmetrically across the entire year. While it is intuitive to assume that thermal deviations would peak alongside the highest absolute temperatures of mid-summer, the data shows a distinct concentration of climate stress during the late-year transition. October logs the most severe shift at 0.0526°C, with September trailing closely at 0.0502°C, whereas the extremes of January and June yield the lowest baselines. This uneven distribution highlights that anomalies are fundamentally transforming transitional shoulder seasons, suggesting that autumn is losing its historical cooling rhythm far more rapidly than other parts of the year.

Twelve-Month Moving Average

A 12-month moving average of the Mean column was calculated separately for each source. The last five rows of the entire DataFrame, which contain the moving averages for the final months of 2016, are presented in Table 4.

Last five rows of the dataset including the moving average:

Source	Date	Raw Mean (°C)	Year	Month	Moving Avg (°C)
GCAG	2016-10-06	0.7292	2016	10	0.9819
GISTEMP	2016-11-06	0.9300	2016	11	1.0175
GCAG	2016-11-06	0.7504	2016	11	0.9640
GISTEMP	2016-12-06	0.8100	2016	12	0.9925
GCAG	2016-12-06	0.7895	2016	12	0.9363

The moving average values are larger than many of the raw monthly values, which reflects the underlying warming trend.

Applying a trailing 12-month smoothing filter to the final stretch of 2016 strips away minor monthly oscillations and exposes the true inertia of the broader climate system. Notably, the moving averages—which edge past the landmark 1.0°C threshold for GISTEMP in November—consistently overshoot the individual raw monthly readings. This happens because the rolling metric carries forward the compounding heat of the preceding months, preventing brief, localized fluctuations from obscuring the larger picture. Ultimately, this structural gap between the raw anomalies and the elevated moving averages confirms that by late 2016, short-term variations were completely overwhelmed by a powerful, sustained upward climate trajectory.

Pivot Table for the Last Twenty Years

A pivot table was built using the df_last20 DataFrame (years 1997 to 2016). The table has years as rows and months as columns. Each cell contains the average temperature anomaly for that specific year-month combination. Table 5 shows a shortened version (first five rows and last five rows). The complete pivot table can be found in the attached Jupyter notebook.

Pivot table of mean temperature anomalies (selected years)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1997	0.352	0.399	0.491	0.418	0.407	0.540	0.430	0.490	0.593	0.644	0.623	0.615
1998	0.607	0.883	0.635	0.688	0.689	0.723	0.723	0.681	0.489	0.483	0.457	0.570
1999	0.490	0.667	0.355	0.400	0.369	0.395	0.406	0.355	0.414	0.404	0.394	0.519
2000	0.303	0.562	0.578	0.611	0.433	0.424	0.406	0.443	0.429	0.301	0.298	0.310
2001	0.456	0.427	0.605	0.543	0.595	0.544	0.584	0.547	0.533	0.510	0.694	0.531
2002	0.723	0.772	0.849	0.576	0.608	0.565	0.621	0.543	0.608	0.524	0.591	0.433
2003	0.710	0.556	0.570	0.555	0.608	0.514	0.545	0.640	0.651	0.739	0.564	0.743
2004	0.594	0.717	0.668	0.607	0.432	0.450	0.370	0.482	0.532	0.639	0.740	0.514
2005	0.671	0.547	0.691	0.711	0.654	0.671	0.659	0.627	0.733	0.747	0.744	0.652
2006	0.513	0.661	0.612	0.509	0.510	0.640	0.581	0.672	0.637	0.684	0.682	0.770
2007	0.922	0.680	0.683	0.747	0.640	0.559	0.574	0.575	0.609	0.584	0.544	0.494
2008	0.257	0.369	0.746	0.506	0.506	0.506	0.590	0.500	0.591	0.670	0.665	0.552
2009	0.608	0.549	0.555	0.632	0.618	0.662	0.682	0.680	0.704	0.641	0.721	0.634
2010	0.713	0.744	0.884	0.849	0.754	0.687	0.674	0.655	0.591	0.668	0.779	0.490
2011	0.490	0.514	0.616	0.651	0.552	0.618	0.679	0.673	0.580	0.647	0.523	0.544
2012	0.441	0.457	0.551	0.712	0.741	0.662	0.629	0.650	0.743	0.741	0.729	0.498
2013	0.634	0.593	0.637	0.541	0.662	0.667	0.628	0.660	0.733	0.684	0.820	0.684
2014	0.712	0.495	0.770	0.790	0.832	0.711	0.634	0.809	0.844	0.818	0.680	0.810
2015	0.812	0.872	0.898	0.757	0.819	0.832	0.759	0.827	0.868	1.026	1.003	1.116
2016	1.113	1.271	1.262	1.082	0.902	0.830	0.849	0.940	0.873	0.810	0.840	0.800

The table clearly shows that temperature anomalies in 2015 and 2016 are considerably higher than those in the late 1990s for almost every month.

Mapping the data across a two-decade matrix exposes a profound generational shift in global climate baselines. When comparing the bookends of this twenty-year span, the thermal deviations of the late 1990s—which typically hovered between 0.3°C and 0.6°C—appear modest against the intense anomalies of 2015 and 2016. During this final two-year stretch, values routinely breach the unprecedented 1.0°C threshold across multiple seasons, peaking dramatically in February 2016 at 1.271°C. This comprehensive monthly breakdown proves that recent warming is not the result of isolated summer spikes; rather, the entire annual cycle has experienced an upward lift, establishing a permanently higher floor for global temperatures across every month of the year.

Machine Learning Results

Only one machine learning model was developed: K-Means clustering on the MNIST digits dataset. A comparative analysis of multiple models is not available. The single model's performance is reported below.

Model: K-Means with 10 clusters (random state = 42)

Dataset: 1,797 samples of handwritten digits (0–9), each flattened to 64 pixel values

Evaluation method: Cluster labels were mapped to the most frequent true digit per cluster. Then the macro-averaged F1 score was computed between the mapped labels and the true labels.

Result:
Macro-averaged F1 score = **0.8628** (approximately 86.3%)

This score indicates a reasonably good alignment between the unsupervised clusters and the actual digit classes. A perfect score of 1.0 would mean every cluster contained only one digit. The obtained value of 0.8628 suggests that most digits were grouped correctly, although some confusion likely occurred between visually similar digits (for example, 1 and 7, or 3 and 8).

No other models (e.g., supervised classifiers, alternative clustering algorithms) were implemented, so a comparison of prediction accuracy levels cannot be provided.

6. Conclusion

The work carried out in this project successfully addressed all the stated objectives. The global temperature dataset was processed by converting the Date column to datetime objects and sorting the entire DataFrame chronologically, which produced a clean time-ordered sequence from 1880 to 2016. A 12-month moving average was calculated separately for the GCAG and GISTEMP sources, and the resulting values for the final months of 2016 ranged from 0.9363 °C to 1.0175 °C (Cahill, Niamh; Rahmstorf, Stefan; Parnell, Andrew C., 2015). These moving averages consistently exceeded the raw monthly readings, confirming that the smoothing operation effectively removed short-term fluctuations and exposed the underlying long-term warming trend. Grouping the data by year showed that yearly average temperatures rose steadily from 0.6295 °C in 2012 to 0.9644 °C in 2016, with no year-to-year decrease during that period. Grouping by month revealed that October had the highest average anomaly (0.0526 °C) across the entire record, followed closely by September (0.0502 °C), indicating that autumn months have experienced the strongest thermal deviations relative to the baseline (NOAA National Centers for Environmental Information, 2024). A pivot table constructed from the last twenty years (1997–2016) clearly demonstrated that temperature anomalies in 2015 and 2016 were substantially higher than those in the late 1990s for almost every month, with February 2016 reaching an anomaly of 1.271 °C (GISTEMP Team, 2024). This confirms that the recent warming is not limited to isolated seasons but has lifted the entire annual cycle.

For the unsupervised learning component, the K-Means clustering model with ten clusters was successfully applied to the MNIST digits dataset. After implementing a custom label mapping routine that assigned each cluster to the most frequent true digit within it, the macro-averaged F1 score was calculated as 0.8628 (Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., 2011). This result shows that even without any labelled training data, the algorithm was able to discover meaningful groupings that largely corresponded to the actual digit classes (Alpaydin & Kaynak, 1998). The performance is reasonable given that no dimensionality reduction or preprocessing was applied to the raw pixel values.

Several recommendations can be made for future extensions of this work. For the temperature time-series analysis, one could incorporate forecasting models such as ARIMA or Prophet to predict near-term temperature anomalies. A statistical test for trend significance, for example the Mann-Kendall test, could be applied to quantify whether the observed warming is statistically significant. Seasonal decomposition using STL (Seasonal-Trend decomposition using Loess) would help separate the trend, seasonal, and residual components more formally. For the clustering part, alternative algorithms such as Agglomerative Clustering, Spectral Clustering, or Gaussian Mixture Models could be evaluated and compared using additional metrics like the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) (Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., 2011). Visualising the cluster separability in two dimensions using PCA or t-SNE would also provide intuitive insight into why certain digits are confused by K-Means.

Appendix A: References

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Appendix B: GitHub Link for Developed Code

All Python code written for this project, including the complete Jupyter Notebook, is available in a public GitHub repository.

GitHub repository link: https://github.com/K-Makorokoto/Time-Series-Analysis-of-Global-Temperature-Data